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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
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Crowd-Sourced Community Infrastructure Damage Mapping with Severity Prioritization

CSA0925-PROGRAMMING IN JAVA

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Problem Background

Application Domain
- Smart City Management
- Infrastructure Monitoring
- Civic Issue Reporting

Current Scenario
- Manual inspections are time-consuming.
- Many issues go unnoticed or are reported late.
- No centralized tracking system.
- Delayed repair and poor resource allocation.

Need for the Project
- Enable citizens to report issues easily.
- Prioritize repairs based on severity.
- Improve response time.
- Increase transparency and accountability.

Target Users
- Citizens
- Municipal Authorities
- Maintenance Teams
- City Administrators

Problem Statement

Problem Definition
- Delayed detection of infrastructure damage.
- Lack of real-time ground-level monitoring.
- Inefficient repair prioritization.

Evidence / Statistics
- Many civic issues remain unreported or unresolved for weeks.
- Manual inspections cover only a small area at a time.
- Delayed repairs increase maintenance costs.

Limitations of Current Systems
- Manual and time-consuming process.
- No centralized reporting platform.
- Poor citizen participation.
- Limited tracking and transparency.

Expected Impact
- Faster issue reporting and response.
- Priority-based repair planning.
- Better resource utilization.
- Improved citizen satisfaction and safer communities.

Objectives

Primary Objective

- Develop a crowd-sourced platform for reporting and prioritizing infrastructure damage.

Measurable Objectives

- Enable geotagged issue reporting by citizens.
- Classify and prioritize damage based on severity.
- Generate heatmaps for quick decision-making.
- Track repair status in real time.
- Improve response time and transparency.

Project Scope

- Citizen issue reporting.
- Severity analysis and heatmap generation.
- Authority task assignment and tracking.
- Status updates for citizens.

Literature Survey

Recent Papers

1. AI-based Road Damage Detection (2024)
2. Smart City Infrastructure Monitoring (2023)
3. Crowd-Sourced Civic Issue Reporting (2023)
4. Deep Learning for Infrastructure Inspection (2024)

Existing Work	Strength	Limitation
AI Road Damage Detection	Accurate detection	No citizen reporting
Smart City Monitoring	Real-time monitoring	Limited public participation
Crowd-Sourced Reporting	Easy issue reporting	No severity prioritization
GIS Damage Mapping	Location-based analysis	No repair tracking

Research Gap

No integrated platform combining citizen reporting, severity-based prioritization, heatmap visualization, and repair status tracking in a single system.

Existing vs Proposed System

Existing System	Proposed System
Manual inspections	Citizen-based reporting
Delayed issue detection	Real-time reporting
No severity prioritization	AI-based severity scoring
Limited tracking	Live repair status tracking

Advantages

- Faster issue identification
- Severity-based repair prioritization
- Improved transparency
- Better resource utilization
- Increased citizen participation

Novelty

- Integrates crowd-sourced reporting, severity scoring, heatmap visualization, and authority action tracking into one smart platform

Requirements

Functional Requirements
- Report infrastructure issues
- Upload geotagged photos
- Generate severity scores & heatmaps
- Track repair status

Non-Functional Requirements
- User-friendly interface
- Secure and reliable
- Scalable performance
- Fast response time

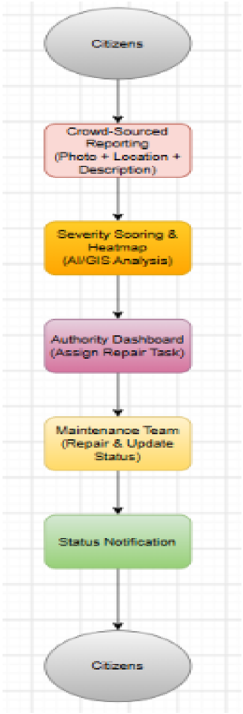
Hardware Requirements
- Computer/Laptop
- Smartphone with GPS & Camera
- Internet connection

Software Requirements
- Frontend: HTML, CSS, JavaScript
- Backend: Java
- Database: MySQL
- Tools: VS Code, Git

Dataset / APIs
- OpenStreetMap / Google Maps API
- GPS Location API
- Municipal infrastructure datasets
- Image dataset for damage classification

Architecture

Block Diagram



Workflow

- Citizen reports an issue.
- System analyzes severity and location.
- Heatmap identifies priority areas.
- Authority assigns repair work.
- Maintenance team updates status.
- Citizen receives progress updates.

Module Description

- **Reporting Module:** Submit geotagged complaints.
- **Severity & Heatmap Module:** Prioritize issues based on impact.
- **Action Tracking Module:** Monitor repair progress and notify users.

Planning

Work Breakdown Structure (WBS)

- Requirement Analysis
- System Design
- Development
- Testing
- Deployment & Documentation

Risk Analysis

Risk	Mitigation
Incorrect reports	Verification & image validation
GPS/Location errors	Accurate geotagging APIs
Low user participation	Simple and user-friendly app
Server downtime	Regular backup & cloud hosting

Expected Outcomes

Deliverables

Web/Mobile reporting application

Severity scoring & heatmap dashboard

Authority action tracking system

Project report and documentation

Benefits

Faster infrastructure issue reporting

Priority-based repair management

Improved transparency and accountability

Better citizen engagement and safer communities

References (IEEE)

1. S. Kumar and R. Singh, “AI-Based Road Damage Detection,” *IEEE Access*, vol. 12, pp. 12345–12356, 2024.
2. J. Wang *et al.*, “Smart City Infrastructure Monitoring Using IoT,” *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4567–4578, 2023.
3. A. Sharma and P. Verma, “Crowd-Sourced Civic Issue Reporting System,” *IEEE Access*, vol. 11, pp. 98765–98776, 2023.

THANK YOU

GPS Photo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